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# Realizations of the double shuffle relations ☆

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ABSTRACT

The set of multiple zeta values satisfy the double shuffle relations. In this paper, we study potential realizations of the double shuffle relations from multiple series and iterated path integrals. We show that the realizations of the double shuffle relations which can be constructed from multiple series are unique up to multiplying by a constant. We prove that there are no nontrivial realizations of the double shuffle relations from iterated path integrals of continuous functions on closed intervals. Lastly we show that the set of one variable multiple polylogarithms satisfies the double shuffle relations only for the point 0 and 1.



## 1. Introduction

Multiple zeta values are defined by

$$\zeta(k_1,\cdots,k_r)=\sum_{0<n_1<\cdots<n_r}\frac{1}{n_1^{k_1}\cdots n_r^{k_r}}, k_1,\cdots,k_{r-1}\geq 1, k_r\geq 2.$$

From the series representations of multiple zeta values, we have the stuffle product on multiple zeta values. For example,

$$\zeta(k_1)\zeta(k_2)=\zeta(k_1,k_2)+\zeta(k_2,k_1)+\zeta(k_1+k_2), k_1,k_2\geq 2.$$

From the iterated integral representations of multiple zeta values, we have the shuffle product on multiple zeta values. For example,

$$\zeta(2)^2=\left(\iint\limits_{0<t_1<t_2<1}\frac{dt_1}{1-t_1}\frac{dt_2}{t_2}\right)^2=2\zeta(2,2)+4\zeta(1,3).$$

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Thus the products of multiple zeta values are $\mathbb{Q}$-linear combinations of multiple zeta values by the double shuffle relations [15].

For $k_r = 1$, the multiple series of the form

$$\sum_{0<n_1<\cdots<n_{r-1}<n_r} \frac{1}{n_1^{k_1}\cdots n_{r-1}^{k_{r-1}} n_r}, k_1, \cdots, k_{r-1} \geq 1$$

are divergent. Ihara, Kaneko and Zagier [15] defined stuffle regularization and shuffle regularization of the above divergent series. They gave the explicit relation between the values of stuffle regularization and shuffle regularization, i.e. the regularized double shuffle relations.

Racinet [17] showed that the models which satisfy the regularized double shuffle relations have a pro-algebraic group structure. We call it the double shuffle group. By the main results of Brown [3], there is a pro-algebraic monomorphism from the pro-unipotent fundamental group of mixed Tate motives over $\mathbb{Z}$ to the double shuffle group. It is widely believed that this monomorphism is also an isomorphism.

Furusho [10] proved that Drinfel'd's pentagon relation in associator theory implies the regularized double shuffle relations. Thus the Gronthendieck-Teichmüller group is contained in the double shuffle group. It is also conjectured that they are isomorphic [11].

The Drinfel'd's rational associator [8] and the Alekseev-Torossian associator [1] both provide realizations of the double shuffle relations, although their coefficients are unrelated to the theory of multiple zeta values. The Deligne's associator [7] which is based on the theory of single valued multiple zeta values also provides a realization of the double shuffle relations.

Gangl, Kaneko and Zagier [12] showed that every $\mathbb{Q}$-coefficients of period polynomial of cusp forms of $SL_2(\mathbb{Z})$ provides a relation among formal double zeta values of the same weight. Goncharov [13] related the double shuffle relations of depth three to the cohomology of modular varieties.

Ma [16] gave a rational realization of formal double zeta space by using the binomial coefficients. Brown [4] and Ecalle [9] gave rational solutions of regularized double shuffle relations up to depth three by using different skills. Bachmann, Kühn and Matthes [2] defined formal double Eisenstein space, which is related to the formal double zeta space and the algebra of quasi-modular forms. For the double shuffle relations of depth two, the structure is quite clear by a series of work. For the double shuffle relations of higher depth, we still know very little.

In this paper, from the theory of multiple series and Chen's iterated path integrals [5], we discuss other possible ways to construct realizations of double shuffle relations.

**Theorem 1.1.** *For a sequence of complex numbers*

$$G = \{a_n \mid n \geq 1\},$$

*assuming that* $\inf\limits_{n\geq 1} \frac{|a_n|}{n} > 0$*, the multiple G-values are defined by*

$$\zeta_G(k_1, \cdots, k_r) = \sum_{0<n_1<\cdots<n_r} \frac{1}{a_{n_1}^{k_1}\cdots a_{n_r}^{k_r}}, k_1, \quad \cdots, k_{r-1} \geq 1, k_r \geq 2.$$

*Then the following two conditions are equivalent:*
*(i) The set of multiple G-values satisfies the double shuffle relations;*
*(ii)* $a_n = cn$, $\forall\, n \geq 1$*, where c is a non-zero complex number.*

Theorem 1.1 shows that the set of multiple zeta values is unique in some sense. Actually we prove a stronger version of Theorem 1.1: The double shuffle relations of depth two are already sufficient to deduce the condition $(ii)$.

Let $F = \{f_0, f_1\}$, where $f_0$ and $f_1$ are continuous $\mathbb{C}$-valued functions on $(0,1]$ and $[0,1)$ respectively. Define the iterated $F$-values by

$$I_F(k_1, \cdots, k_r) = \int_{0<t_1<\cdots<t_k<1} \cdots \int f_{i_1}(t_1) \cdots f_{i_k}(t_k) dt_1 \cdots dt_k,$$

where $k_1, \cdots, k_{r-1} \geq 1, k_r \geq 2$, $k = k_1 + \cdots + k_r$ and

$$f_{i_j}(t) = \begin{cases} f_1(t), & j \in \{1, 1+k_1, \cdots, 1+k_1+\cdots+k_{r-1}\}; \\ f_0(t), & j \notin \{1, 1+k_1, \cdots, 1+k_1+\cdots+k_{r-1}\}. \end{cases}$$

If $f_0(t) = \frac{1}{t}, f_1(t) = \frac{1}{1-t}$, then it is clear that

$$I_F(k_1, \cdots, k_r) = \zeta(k_1, \cdots, k_r).$$

**Theorem 1.2.** *Assuming that $f_0$ and $f_1$ are continuous functions on $[0,1]$, if the set of iterated $F$-values satisfies the double shuffle relations, then all the iterated $F$-values are trivial, i.e.*

$$I_F(k_1, \cdots, k_r) = 0,\ k_1, \cdots, k_{r-1} \geq 1, k_r \geq 2.$$

Theorem 1.2 shows that in order to find a non-trivial model of double shuffle relations, one needs to consider unbounded functions.

The one variable multiple polylogarithms are defined by

$$\mathrm{Li}_{k_1,\cdots,k_r}(z) = \sum_{1 \leq n_1 < \cdots < n_r} \frac{z^{n_r}}{n_1^{k_1} \cdots n_r^{k_r}}, |\, z\,| < 1,\ k_1, \cdots, k_r \geq 1.$$

The set of one variable multiple polylogarithms satisfies the shuffle product formula for any $|z| \leq 1$. It is clear that the set of one variable multiple polylogarithms

$$\left\{ \mathrm{Li}_{k_1,\cdots,k_r}(z) \,\middle|\, k_1, \cdots, k_{r-1} \geq 1, k_r \geq 2 \right\}$$

satisfies the double shuffle relations for $z = 0$ (the trivial realization) and for $z = 1$ (realization of multiple zeta values). The following theorem shows that there are no similar points besides 0 and 1.

**Theorem 1.3.** *For $|z| \leq 1$, if the set of one variable multiple polylogarithms*

$$\left\{ \mathrm{Li}_{k_1,\cdots,k_r}(z) \,\middle|\, k_1, \cdots, k_{r-1} \geq 1, k_r \geq 2 \right\}$$

*satisfies the double shuffle relations, then $z = 0$ or $1$.*

## 2. Regularized double shuffle relations

In this section we review the theory of regularized double shuffle relations applied to multiple zeta values. The main reference is [15].

For integers $n_1, \cdots, n_r \geq 1$, the multiple series

$$\sum_{1<m_1<\cdots<m_r} \frac{1}{m_1^{n_1} \cdots m_r^{n_r}}$$

is convergent if and only if $n_r \geq 2$. For $n_r = 1$, it is divergent.

However, there are two reasonable ways to define the value of $\zeta(n_1, \cdots, n_r)$ in the case $n_r = 1$. For $n_1, \cdots, n_r \geq 1$ and $N$ is big enough, it is easy to check that

$$\sum_{0<m_1<\cdots<m_r<N} \frac{1}{m_1^{n_1} \cdots m_r^{n_r}} = p_{n_1,\cdots,n_r}(\log N + \gamma) + O\left(\frac{(\log N)^{J_1}}{N}\right).$$

Here $J_1 \geq 0$, $p(T) \in \mathbb{R}[T]$ and all the coefficients of $p(T)$ belong to $\mathcal{Z}$. The stuffle regularized multiple zeta values are defined by

$$\zeta^{st}(n_1, \cdots, n_r) = p_{n_1,\cdots,n_r}(T).$$

It is clear that $\zeta^{st}(n_1, \cdots, n_r) = \zeta(n_1, \cdots, n_r)$ for $n_r \geq 2$.

Similarly, for any $k_1, \cdots, k_r \geq 1$, one can check that

$$\int \cdots \int_{0<t_1<\cdots<t_K<1-\epsilon} \omega_{i_1}(t_1) \cdots \omega_{i_K}(t_K) = q_{n_1,\cdots,n_r}(\log \epsilon) + O\left(\epsilon \log^{J_2} \epsilon\right).$$

Here $J_2 \geq 0$, $q(T) \in \mathbb{R}[T]$ and all the coefficients of $q(T)$ belong to $\mathcal{Z}$. The shuffle regularized multiple zeta values are defined by

$$\zeta^{sh}(n_1, \cdots, n_r) = q_{n_1,\cdots,n_r}(T).$$

It is also clear that $\zeta^{sh}(n_1, \cdots, n_r) = \zeta(n_1, \cdots, n_r)$ for $n_r \geq 2$.

Denote by $\mathfrak{h} = \mathbb{Q}\langle x, y\rangle$ the non-commutative polynomial ring over $\mathbb{Q}$ in two indeterminants $x$ and $y$. Define $\mathfrak{h}^0 := \mathbb{Q} + y\mathfrak{h}x$ and $\mathfrak{h}^1 := \mathbb{Q} + y\mathfrak{h}$. Denote by

$$Y = \mathbb{Q}\langle y_1, y_2, \cdots, y_n, \cdots\rangle$$

the non-commutative polynomial ring over $\mathbb{Q}$ in indeterminants $y_1, y_2, \cdots, y_n, \cdots$. There is a natural isomorphism $\tau$ between $\mathfrak{h}^1$ and $Y$:

$$\tau : \mathfrak{h}^1 \to Y,$$
$$yx^{n_1-1}yx^{n_2-1} \cdots yx^{n_r-1} \mapsto y_{n_1}y_{n_2} \cdots y_{n_r}.$$

Thus one can identify $\mathfrak{h}^1$ with $Y$. For $y_{n_1}y_{n_2} \cdots y_{n_r}$, $r$ and $n_1 + n_2 + \cdots + n_r$ are called its depth and weight respectively. Denote by

$$wt(y_{n_1}y_{n_2} \cdots y_{n_r}) = n_1 + n_2 + \cdots + n_r$$

and $dep(y_{n_1}y_{n_2} \cdots y_{n_r}) = r$.

Now we define two new products on $\mathfrak{h}^1$. The stuffle product $*$ on $\mathfrak{h}^1$ is defined inductively by

$$1 * u = u * 1 = u,$$
$$y_k u * y_l v = y_k(u * y_l v) + y_l(y_k u * v) + y_{k+l}(u * v).$$

The shuffle product $\sqcup\!\sqcup$ on $\mathfrak{h}^1$ is defined inductively by

$$1 \sqcup\!\sqcup u = u \sqcup\!\sqcup 1 = u,$$

$$u_1u \sqcup\!\sqcup v_1v = u_1(u \sqcup\!\sqcup v_1v) + v_1(u_1u \sqcup\!\sqcup v),$$

where $u_1, v_1 \in \{x, y\}$. The two products are both commutative.

The theory of regularized double shuffle relations is summarized in the following theorem.

**Theorem 2.1.** *(Ihara-Kaneko-Zagier) Define $Z^{st} : \mathfrak{h}^1 \to \mathcal{Z}[T]$ as*

$$y_{k_1}y_{k_2}\cdots y_{k_r} \mapsto \zeta^{st}(k_1, k_2, \cdots, k_r).$$

*and $Z^{sh} : \mathfrak{h}^1 \to \mathcal{Z}[T]$ as*

$$y_{k_1}y_{k_2}\cdots y_{k_r} \mapsto \zeta^{sh}(k_1, k_2, \cdots, k_r).$$

*Then we have:*
*(i) $Z^{st}(u * v) = Z^{st}(u)Z^{st}(v)$;*
*(ii) $Z^{sh}(u \sqcup\!\sqcup v) = Z^{sh}(u)Z^{sh}(v)$;*
*(iii) $\rho \circ Z^{st} = Z^{sh}$, where $\rho : \mathbb{R}[T] \to \mathbb{R}[T]$ is the linear map defined by*

$$\rho(e^{Tu}) = A(u)e^{Tu}, A(u) = \exp\left(\sum_{n\geq 2} \frac{(-1)^n}{n}\zeta(n)u^n\right).$$

## 3. From multiple series to double shuffle relations

For $G = \{a_n \mid n \geq 1\}$, from the condition $\inf_{n\geq 1} \frac{|a_n|}{n} > 0$, it is clear that the multiple $G$-value

$$\zeta_G(k_1, \cdots, k_r) = \sum_{0<n_1<\cdots<n_r} \frac{1}{a_{n_1}^{k_1}\cdots a_{n_r}^{k_r}}$$

is convergent if $k_1, \cdots, k_{r-1} \geq 1, k_r \geq 2$. Thus one can define the map

$$Z_G : \mathfrak{h}^0 \to \mathbb{C}$$

as

$$yx^{k_1-1}yx^{k_2-1}\cdots yx^{k_r-1} \mapsto \zeta_G(k_1, k_2, \cdots, k_r).$$

From the construction of multiple $G$-values, it is obvious that

$$Z_G(u * v) = Z_G(u)Z_G(v), \forall\, u, v \in \mathfrak{h}^0.$$

Multiple $G$-values have some similarities to multiple zeta values. For example, multiple $G$-values also satisfy Theorem 2.2 in [14].

For the shuffle product, one has

**Lemma 3.1.** *For the set of multiple $G$-values, define*

$$\mathfrak{z}_G(s) = \sum_{k\geq 2} \zeta_G(k)s^{k-1},$$

$$\mathfrak{Z}_G(s_1,s_2)=\sum_{\substack{k_1+k_2\geq 4\\ k_1\geq 1,k_2\geq 2}}\zeta_G(k_1,k_2)s_1^{k_1-1}s_2^{k_2-1}.$$

*Then the following two conditions are equivalent:*
*(i)* $Z_G(yx^{k_1-1}\sqcup\!\sqcup\, yx^{k_2-1})=\zeta_G(k_1)\zeta_G(k_2),\ \forall\, k_1,k_2\geq 2$;
*(ii)*

$$\begin{aligned}&\mathfrak{Z}_G(s_1)\mathfrak{Z}_G(s_2)\\ &=\mathfrak{Z}_G(s_1,s_1+s_2)+\mathfrak{Z}_G(s_2,s_1+s_2)-\mathfrak{Z}_G(s_1,s_1)-\mathfrak{Z}_G(s_2,s_2)-\mathfrak{Z}_G(0,s_1)-\mathfrak{Z}_G(0,s_2).\end{aligned}$$

**Proof.** Define $\binom{m}{l}=\frac{m!}{l!(m-l)!}$, if $m,l\geq 0, l\leq m$ and $\binom{m}{l}=0$, if $m,l\geq 0, l>m$. Define

$$\delta_{a,b}=\begin{cases}1 & a=b;\\ 0 & a\neq b.\end{cases}$$

By induction, one can check that

$$x^a\sqcup\!\sqcup\, x^b=\binom{a+b}{a}x^{a+b},\ a,b\geq 0.$$

Thus

$$\begin{aligned}&yx^{a-1}\sqcup\!\sqcup\, x^{b-1}\\ &=y(x^{a-1}\sqcup\!\sqcup\, x^{b-1})+x(yx^{a-1}\sqcup\!\sqcup\, x^{b-2})\\ &=\qquad\cdots\\ &=\sum_{r=0}^{b-1}x^ry(x^{a-1}\sqcup\!\sqcup\, x^{b-1-r})\\ &=\sum_{r=0}^{b-1}\binom{a+b-2-r}{a-1}x^ryx^{a+b-2-r}.\end{aligned}$$

As a result, for $k_1,k_2\geq 2$, one has

$$\begin{aligned}&yx^{k_1-1}\sqcup\!\sqcup\, yx^{k_2-1}\\ &=y(x^{k_1-1}\sqcup\!\sqcup\, yx^{k_2-1})+y(yx^{k_1-1}\sqcup\!\sqcup\, x^{k_2-1})\\ &=\sum_{\substack{l_1+l_2=k_1+k_2\\ l_1\geq 1,l_2\geq 2}}\left[\binom{l_2-1}{k_1-1}+\binom{l_2-1}{k_2-1}\right]yx^{l_1-1}yx^{l_2-1}.\end{aligned}$$

It follows that $(i)$ is equivalent to

$$\zeta_G(k_1)\zeta_G(k_2)=\sum_{\substack{l_1+l_2=k_1+k_2\\ l_1\geq 1,l_2\geq 2}}\left[\binom{l_2-1}{k_1-1}+\binom{l_2-1}{k_2-1}\right]\zeta_G(l_1,l_2),\ \forall\, k_1,k_2\geq 2$$

By multiplying $s_1^{k_1-1}s_2^{k_2-1}$ to the above formula for $k_1,k_2\geq 2$, one has that the condition $(i)$ is equivalent to

$$
\begin{aligned}
&\mathfrak{Z}_G(s_1)\mathfrak{Z}_G(s_2)\\
&=\sum_{k_1,k_2\geq 2}\left\{\sum_{\substack{l_1+l_2=k_1+k_2\\ l_1\geq 1, l_2\geq 2}}\left[\binom{l_2-1}{k_1-1}+\binom{l_2-1}{k_2-1}\right]\zeta_G(l_1,l_2)\right\}s_1^{k_1-1}s_2^{k_2-1}\\
&=\sum_{\substack{l_1+l_2\geq 4\\ l_1\geq 1, l_2\geq 2}}\zeta_G(l_1,l_2)\left\{\sum_{\substack{k_1+k_2=l_1+l_2\\ k_1,k_2\geq 2}}\left[\binom{l_2-1}{k_1-1}+\binom{l_2-1}{k_2-1}\right]s_1^{k_1-1}s_2^{k_2-1}\right\}\\
&=\sum_{\substack{l_1+l_2\geq 4\\ l_1\geq 1, l_2\geq 2}}\zeta_G(l_1,l_2)\\
&\quad\cdot\left[(s_1^{l_1-1}+s_2^{l_1-1})(s_1+s_2)^{l_2-1}-s_1^{l_1+l_2-2}-s_2^{l_1+l_2-2}-\delta_{l_1,1}\left(s_1^{l_2-1}+s_2^{l_2-1}\right)\right]\\
&=\mathfrak{Z}_G(s_1,s_1+s_2)+\mathfrak{Z}_G(s_2,s_1+s_2)-\mathfrak{Z}_G(s_1,s_1)-\mathfrak{Z}_G(s_2,s_2)-\mathfrak{Z}_G(0,s_1)-\mathfrak{Z}_G(0,s_2). \quad\square
\end{aligned}
$$

For the analytic properties of $\mathfrak{Z}_G(s)$ and $\mathfrak{Z}_G(s_1,s_2)$, one has

**Lemma 3.2.** *For $G=\{a_n \mid n\geq 1\}$ which satisfies the condition $\inf\limits_{n\geq 1}\frac{|a_n|}{n}>0$, we have*
*(i) $\mathfrak{Z}_G(s)$ is a meromorphic function over $\mathbb{C}$ and*

$$\mathfrak{Z}_G(s)=\sum_{n\geq 1}\frac{s}{a_n(a_n-s)};$$

*(ii) $\mathfrak{Z}_G(s_1,s_2)$ is a meromorphic function over $\mathbb{C}^2$ and*

$$\mathfrak{Z}_G(s_1,s_2)=\sum_{0<n_1<n_2}\frac{s_2}{a_{n_2}}\left[\frac{1}{(a_{n_1}-s_1)(a_{n_2}-s_2)}-\frac{1}{a_{n_1}a_{n_2}}\right];$$

**Proof.** $(i)$ It is clear that

$$
\begin{aligned}
&\mathfrak{Z}_G(s)\\
&=\sum_{k\geq 2}\zeta_G(k)s^{k-1}=\frac{1}{s}\sum_{n\geq 1,k\geq 2}\left(\frac{s}{a_n}\right)^k=\frac{1}{s}\sum_{n\geq 1}\frac{s^2}{a_n^2}\frac{1}{1-\frac{s}{a_n}}=\sum_{n\geq 1}\frac{s}{a_n(a_n-s)}.
\end{aligned}
$$

Since $|a_n|\geq An$, $n\geq 1$, for some $A>0$, the series

$$\sum_{n\geq 1}\frac{s}{a_n(a_n-s)}$$

is absolutely convergent for any $s\notin G$. Thus $\mathfrak{Z}_G(s)$ is a meromorphic function over $\mathbb{C}$. By exactly the same way, one can prove the statement $(ii)$. $\square$

The following lemma will be used in this section.

**Lemma 3.3.** *For $G=\{a_n \mid n\geq 1\}$ which satisfies*

$$a_n\neq 0, \forall\, n\geq 1,$$

*if*

$$\{a_1, a_2, \cdots, a_m, \cdots\}$$

*and*

$$\{a_{n+1} - a_n, a_{n+2} - a_n, \cdots, a_{n+m} - a_n, \cdots\}$$

*are the same as sets for every $n \geq 1$, then $a_n = cn$ for some non-zero $c$.*

**Proof.** If $a_2 - a_1 = a_i$ for some $i$, there are three cases:
(1) $i = 1$, $a_2 = 2a_1$.
(2) $i = 2$, we will have $a_1 = 0$. Thus this is impossible.
(3) $i \geq 3$, we will have $a_i - a_2 = -a_1$. So

$$-a_1 \in \{a_3 - a_2, \cdots, a_m - a_2, \cdots\} = \{a_1, a_2, \cdots, a_m, \cdots\} = \{a_2 - a_1, \cdots, a_m - a_1, \cdots\}.$$

Then $-a_1 = a_{i'} - a_1$ for some $i' > 1$. This shows that $a_{i'} = 0$, which is also impossible. In conclusion, we have $a_2 = 2a_1$.

By induction, assuming that $a_n = na_1$ for $n \leq N$, since

$$\{a_2 - a_1, \cdots, a_m - a_1, \cdots\} = \{a_1, a_2, \cdots, a_m, \cdots\},$$

it follows that

$$a_{N+1} - a_1 = a_j$$

for some $j$.
$(A)$ If $1 \leq j \leq N - 1$, then

$$a_{N+1} = a_1 + ja_1 = (j + 1)a_1 = a_{j+1}.$$

It is impossible since for $1 \leq j \leq N - 1$,

$$a_{N+1} - a_{j+1} \in \{a_1, \cdots, a_m, \cdots\}$$

and $a_n \neq 0, \forall n \geq 1$.
$(B)$ If $j = N$, $a_{N+1} = (N + 1)a_1$.
$(C)$ If $j = N + 1$, we will have $a_1 = 0$, which is impossible.
$(D)$ If $j > N + 1$, then $-a_1 = a_j - a_{N+1} \in \{a_2 - a_1, \cdots, a_m - a_1, \cdots\}$. Thus $-a_1 = a_l - a_1$ for some $l \geq 2$. It follows that $a_l = 0$. We have a contradiction here.

From the above analysis, $a_n = cn, \forall\, n \geq 1$. □

Now we are ready to prove our main result.

**Theorem 3.4.** *For $G = \{a_n \mid n \geq 1\}$ which satisfies the condition $\inf\limits_{n\geq 1} \frac{|a_n|}{n} > 0$, the following two statements are equivalent:*
*(i) $a_n = cn, n \geq 1$, for some non-zero complex number $c$;*
*(ii) $Z_G(yx^{k_1-1} ⧢ yx^{k_2-1}) = \zeta_G(k_1)\zeta_G(k_2)$, $\forall\, k_1, k_2 \geq 2$.*

**Proof.** From $(i)$ to $(ii)$, one can use the fact that the set of multiple zeta values satisfies the shuffle product formula and the shuffle product formula is graded by weight. But here we prefer to give a different approach since this approach gives us a different way to understand the shuffle product.

By Lemma 3.1 and Lemma 3.2, the statement $(ii)$ is equivalent to

$$\begin{aligned}
&\sum_{n_1,n_2\geq 1}\frac{s_1s_2}{a_{n_1}(a_{n_1}-s_1)a_{n_2}(a_{n_2}-s_2)}\\
&=\sum_{0<n_1<n_2}\frac{s_1+s_2}{a_{n_2}}\left[\frac{1}{(a_{n_1}-s_1)(a_{n_2}-s_1-s_2)}+\frac{1}{(a_{n_1}-s_2)(a_{n_2}-s_1-s_2)}-\frac{2}{a_{n_1}a_{n_2}}\right]\\
&-\sum_{0<n_1<n_2}\left\{\frac{s_1}{a_{n_2}}\left[\frac{1}{(a_{n_1}-s_1)(a_{n_2}-s_1)}-\frac{1}{a_{n_1}a_{n_2}}\right]+\frac{s_2}{a_{n_2}}\left[\frac{1}{(a_{n_1}-s_2)(a_{n_2}-s_2)}-\frac{1}{a_{n_1}a_{n_2}}\right]\right\}\\
&-\sum_{0<n_1<n_2}\left\{\frac{s_1}{a_{n_2}}\left[\frac{1}{a_{n_1}(a_{n_2}-s_1)}-\frac{1}{a_{n_1}a_{n_2}}\right]+\frac{s_2}{a_{n_2}}\left[\frac{1}{a_{n_1}(a_{n_2}-s_2)}-\frac{1}{a_{n_1}a_{n_2}}\right]\right\}.
\end{aligned}$$

One can rewrite the above equation as

$$\begin{aligned}
&\sum_{n_1,n_2\geq 1}\frac{s_1s_2}{a_{n_1}a_{n_2}(a_{n_1}-s_1)(a_{n_2}-s_2)}\\
&=\sum_{0<n_1<n_2}\frac{s_1+s_2}{a_{n_2}}\left[\frac{1}{(a_{n_1}-s_1)(a_{n_2}-s_1-s_2)}+\frac{1}{(a_{n_1}-s_2)(a_{n_2}-s_1-s_2)}\right]\\
&-\sum_{0<n_1<n_2}\left[\frac{s_1}{a_{n_2}}\cdot\frac{1}{(a_{n_1}-s_1)(a_{n_2}-s_1)}+\frac{s_2}{a_{n_2}}\cdot\frac{1}{(a_{n_1}-s_2)(a_{n_2}-s_2)}\right]\\
&-\sum_{0<n_1<n_2}\left[\frac{s_1}{a_{n_2}}\cdot\frac{1}{a_{n_1}(a_{n_2}-s_1)}+\frac{s_2}{a_{n_2}}\cdot\frac{1}{a_{n_1}(a_{n_2}-s_2)}\right]
\end{aligned}$$

$$\begin{aligned}
=&\sum_{0<n_1<n_2}\frac{s_1}{a_{n_2}}\left[\frac{1}{(a_{n_1}-s_1)(a_{n_2}-s_1-s_2)}+\frac{1}{(a_{n_1}-s_2)(a_{n_2}-s_1-s_2)}\right.\\
&\qquad\left.-\frac{1}{(a_{n_1}-s_1)(a_{n_2}-s_1)}-\frac{1}{a_{n_1}(a_{n_2}-s_1)}\right]\\
&+\sum_{0<n_1<n_2}\frac{s_2}{a_{n_2}}\left[\frac{1}{(a_{n_1}-s_1)(a_{n_2}-s_1-s_2)}+\frac{1}{(a_{n_1}-s_2)(a_{n_2}-s_1-s_2)}\right.\\
&\qquad\left.-\frac{1}{(a_{n_1}-s_2)(a_{n_2}-s_2)}-\frac{1}{a_{n_1}(a_{n_2}-s_2)}\right]
\end{aligned}\tag{1}$$

If $a_n=cn,\forall\, n\geq 1$ and $c\neq 0$, then by the following observation

$$\frac{1}{uv}=\frac{1}{u(u+v)}+\frac{1}{v(u+v)},$$

one has

$$\begin{aligned}
&\sum_{0<n_1<n_2} \frac{s_1}{cn_2}\Bigg[\frac{1}{(cn_1-s_1)(cn_2-s_1-s_2)}+\frac{1}{(cn_1-s_2)(cn_2-s_1-s_2)}\\
&\qquad -\frac{1}{(cn_1-s_1)(cn_2-s_1)}-\frac{1}{cn_1(cn_2-s_1)}\Bigg]\\
&+\sum_{0<n_1<n_2} \frac{s_2}{cn_2}\Bigg[\frac{1}{(cn_1-s_1)(cn_2-s_1-s_2)}+\frac{1}{(cn_1-s_2)(cn_2-s_1-s_2)}\\
&\qquad -\frac{1}{(cn_1-s_2)(cn_2-s_2)}-\frac{1}{cn_1(cn_2-s_2)}\Bigg]\\
&=\sum_{n_1,n_2\geq 1} \frac{s_1}{cn_1+cn_2}\Bigg[\frac{1}{(cn_1-s_1)(cn_1+cn_2-s_1-s_2)}+\frac{1}{(cn_1-s_2)(cn_1+cn_2-s_1-s_2)}\\
&\qquad -\frac{1}{(cn_1-s_1)(cn_1+cn_2-s_1)}-\frac{1}{cn_1(cn_1+cn_2-s_1)}\Bigg]\\
&+\sum_{n_1,n_2\geq 1} \frac{s_2}{cn_1+cn_2}\Bigg[\frac{1}{(cn_1-s_1)(cn_1+cn_2-s_1-s_2)}+\frac{1}{(cn_1-s_2)(cn_1+cn_2-s_1-s_2)}\\
&\qquad -\frac{1}{(cn_1-s_2)(cn_1+cn_2-s_2)}-\frac{1}{cn_1(cn_1+cn_2-s_2)}\Bigg]\\
&=\sum_{n_1,n_2\geq 1} \frac{s_1}{cn_1+cn_2}\Bigg[\frac{1}{(cn_1-s_1)(cn_1+cn_2-s_1-s_2)}+\frac{1}{(cn_2-s_2)(cn_1+cn_2-s_1-s_2)}\\
&\qquad -\frac{1}{(cn_1-s_1)(cn_1+cn_2-s_1)}-\frac{1}{cn_2(cn_1+cn_2-s_1)}\Bigg]\\
&+\sum_{n_1,n_2\geq 1} \frac{s_2}{cn_1+cn_2}\Bigg[\frac{1}{(a_{n_2}-s_1)(a_{n_1}+a_{n_2}-s_1-s_2)}+\frac{1}{(a_{n_1}-s_2)(a_{n_1}+a_{n_2}-s_1-s_2)}\\
&\qquad -\frac{1}{(cn_1-s_2)(cn_1+cn_2-s_2)}-\frac{1}{cn_2(cn_1+cn_2-s_2)}\Bigg]\\
&=\sum_{n_1,n_2\geq 1} \frac{s_1}{cn_1+cn_2}\left[\frac{1}{(cn_1-s_1)(cn_2-s_2)}-\frac{1}{(cn_1-s_1)cn_2}\right]\\
&\qquad +\sum_{n_1,n_2\geq 1} \frac{s_2}{cn_1+cn_2}\left[\frac{1}{(cn_1-s_2)(cn_2-s_1)}-\frac{1}{(cn_1-s_2)cn_2}\right]\\
&=\sum_{n_1,n_2\geq 1}\left[\frac{s_1s_2}{cn_2(cn_1+cn_2)(cn_1-s_1)(cn_2-s_2)}+\frac{s_1s_2}{cn_2(cn_1+cn_2)(cn_1-s_2)(cn_2-s_1)}\right]\\
&=\sum_{n_1,n_2\geq 1}\left[\frac{s_1s_2}{cn_2(cn_1+cn_2)(cn_1-s_1)(cn_2-s_2)}+\frac{s_1s_2}{cn_1(cn_1+cn_2)(cn_2-s_2)(cn_1-s_1)}\right]\\
&=\sum_{n_1,n_2\geq 1}\frac{s_1s_2}{c^2n_1n_2(cn_1-s_1)(cn_2-s_2)}.
\end{aligned}$$

In the second equality, we use the fact that: for $w = s_1$ or $s_2$,

$$\sum_{n_1,n_2\geq 1}\frac{w}{cn_1+cn_2}\cdot\frac{1}{cn_1(cn_1+cn_2-s_1)}=\sum_{n_1,n_2\geq 1}\frac{w}{cn_1+cn_2}\cdot\frac{1}{cn_2(cn_1+cn_2-s_1)}.$$

In the above identity, we only replace the term $cn_1$ on the left side by the term $cn_2$ on the right side. We use the same observation in the fifth equality.

For $(ii) \Rightarrow (i)$, by Lemma 3.1 and Lemma 3.2, one has the formula (1) as an equation of meromorphic functions over $\mathbb{C}^2$. Firstly we will show that

$$a_m \neq a_n, \forall\, m \neq n.$$

By the condition $\inf\limits_{n\geq 1} \frac{|a_n|}{n} > 0$, the set

$$\{m \mid a_m = a_n\}$$

is a finite set for $n \geq 1$. If there is an $n$ which satisfies

$$r_n = \#\{m \mid a_m = a_n\} \geq 2,$$

then

$$\lim_{s_2 \to a_n} (a_n - s_2)^2 \sum_{n_1, n_2 \geq 1} \frac{s_1 s_2}{a_{n_1} a_{n_2} (a_{n_1} - s_1)(a_{n_2} - s_2)} = 0$$

and

$$\begin{aligned}
&\lim_{s_2 \to a_n} (a_n - s_2)^2 \sum_{0<n_1<n_2} \left\{ \frac{s_1}{a_{n_2}} \left[ \frac{1}{(a_{n_1} - s_1)(a_{n_2} - s_1 - s_2)} \right.\right. \\
&+ \left. \frac{1}{(a_{n_1} - s_2)(a_{n_2} - s_1 - s_2)} - \frac{1}{(a_{n_1} - s_1)(a_{n_2} - s_1)} - \frac{1}{a_{n_1}(a_{n_2} - s_1)} \right] \\
&+ \frac{s_2}{a_{n_2}} \left[ \frac{1}{(a_{n_1} - s_1)(a_{n_2} - s_1 - s_2)} + \frac{1}{(a_{n_1} - s_2)(a_{n_2} - s_1 - s_2)} - \frac{1}{(a_{n_1} - s_2)(a_{n_2} - s_2)} \right. \\
&- \left.\left. \frac{1}{a_{n_1}(a_{n_2} - s_2)} \right] \right\} \\
&\qquad = -\lim_{s_2 \to a_n} (a_n - s_2)^2 \sum_{0<n_1<n_2} \frac{s_2}{a_{n_2}} \cdot \frac{1}{(a_{n_1} - s_2)(a_{n_2} - s_2)} = -\binom{r_n}{2}.
\end{aligned}$$

The above two limits contradict the formula (1). So we have

$$a_m \neq a_n, \forall\, m \neq n.$$

For $n \geq 1$,

$$\lim_{s_2 \to a_n} (a_n - s_2) \sum_{n_1, n_2 \geq 1} \frac{s_1 s_2}{a_{n_1} a_{n_2} (a_{n_1} - s_1)(a_{n_2} - s_2)} = \sum_{n_1 \geq 1} \frac{s_1}{a_{n_1}(a_{n_1} - s_1)}.$$

For any $n \geq 1$, one has

$$\begin{aligned}
&\lim_{s_2\to a_n}(a_n-s_2)\sum_{0<n_1<n_2}\Bigg\{\frac{s_1}{a_{n_2}}\Bigg[\frac{1}{(a_{n_1}-s_1)(a_{n_2}-s_1-s_2)}\\
&+\frac{1}{(a_{n_1}-s_2)(a_{n_2}-s_1-s_2)}-\frac{1}{(a_{n_1}-s_1)(a_{n_2}-s_1)}-\frac{1}{a_{n_1}(a_{n_2}-s_1)}\Bigg]\\
&+\frac{s_2}{a_{n_2}}\Bigg[\frac{1}{(a_{n_1}-s_1)(a_{n_2}-s_1-s_2)}+\frac{1}{(a_{n_1}-s_2)(a_{n_2}-s_1-s_2)}-\frac{1}{(a_{n_1}-s_2)(a_{n_2}-s_2)}\\
&-\frac{1}{a_{n_1}(a_{n_2}-s_2)}\Bigg]\Bigg\}\\
&=\sum_{n_2>n}\left[\frac{s_1}{a_{n_2}}\cdot\frac{1}{a_{n_2}-a_n-s_1}+\frac{a_n}{a_{n_2}}\left(\frac{1}{a_{n_2}-a_n-s_1}-\frac{1}{a_{n_2}-a_n}\right)\right]\\
&\qquad\qquad-\sum_{0<n_1<n}\left(\frac{1}{a_{n_1}-a_n}+\frac{1}{a_{n_1}}\right)\\
&=\sum_{n_2>n}\left[\frac{s_1+a_n}{a_{n_2}}\cdot\frac{1}{a_{n_2}-a_n-s_1}-\frac{a_n}{a_{n_2}}\cdot\frac{1}{a_{n_2}-a_n}\right]-\sum_{0<n_1<n}\left(\frac{1}{a_{n_1}-a_n}+\frac{1}{a_{n_1}}\right).
\end{aligned}$$

Thus by the formula (1), for any $n\geq 1$, we have

$$\begin{aligned}
&\sum_{n_1\geq 1}\frac{s_1}{a_{n_1}(a_{n_1}-s_1)}\\
&=\sum_{n_2>n}\left[\frac{s_1+a_n}{a_{n_2}}\cdot\frac{1}{a_{n_2}-a_n-s_1}-\frac{a_n}{a_{n_2}}\cdot\frac{1}{a_{n_2}-a_n}\right]-\sum_{0<n_1<n}\left(\frac{1}{a_{n_1}-a_n}+\frac{1}{a_{n_1}}\right).
\end{aligned}$$

By comparing the poles of the meromorphic functions in the above formula, it follows that the two sets

$$\{a_1,a_2,\cdots,a_m,\cdots\}$$

and

$$\{a_{n+1}-a_n,a_{n+2}-a_n,\cdots,a_{n+m}-a_n,\cdots\}$$

are the same for any $n\geq 1$. By Lemma 3.3, the theorem is proved. □

Theorem 1.1 follows from Theorem 3.4 immediately.

**Remark 3.5.** In Theorem 1.1 and Theorem 3.4, the condition

$$\inf_{n\geq 1}\frac{|a_n|}{n}>0$$

can be replaced by the following condition: The multiple series

$$\zeta_G(\underbrace{1,\cdots,1}_{r},2)$$

is absolutely convergent for any $r\geq 0$.

## 4. From iterated integrals to double shuffle relations

For $F = (f_0, f_1)$, where $f_0$ and $f_1$ are continuous $\mathbb{C}$-valued functions on $(0,1]$ and $[0,1)$ respectively, one can construct the following iterated $F$-values:

$$I_F(k_1, \cdots, k_r) = \int \cdots \int_{0<t_1<\cdots<t_K<1} f_{i_1}(t_1) \cdots f_{i_K}(t_K) dt_1 \cdots dt_K,$$

where $k_1, \cdots, k_{r-1} \geq 1, k_r \geq 2$ and

$$f_{i_j}(t) = \begin{cases} f_1(t), & j \in \{1, 1+i_1, \cdots, 1+i_1+\cdots+i_{r-1}\}; \\ f_0(t), & j \notin \{1, 1+i_1, \cdots, 1+i_1+\cdots+i_{r-1}\}. \end{cases}$$

Assuming that

$$|f_1(t)| < \frac{M}{1-t},\ |f_0(t)| < \frac{M}{t}, \forall\, t \in (0,1),$$

by the iterated integral representations of multiple zeta values it follows that the iterated $F$-value

$$I_F(k_1, \cdots, k_r)$$

is convergent for $k_1, \cdots, k_{r-1} \geq 1, k_r \geq 2$.

**Remark 4.1.** In general cases, the condition

$$|f_1(t)| < \frac{M}{1-t},\ |f_0(t)| < \frac{M}{t}, \forall\, t \in (0,1)$$

is not necessary. We only need the following condition: The iterated $F$-value

$$I_F(k_1, \cdots, k_r)$$

is convergent for every $k_1, \cdots, k_{r-1} \geq 1, k_r \geq 2$.

We wish to find a set of iterated $F$-values which satisfies double shuffle relations. Here we need a lemma about the stuffle product $*$ on $\mathfrak{h}^1 \cong Y$.

**Lemma 4.2.** *Denote by* $\mathrm{E} : \mathfrak{h}^1 \to \mathbb{Q}$ *the* $\mathbb{Q}$*-linear map which satisfies*

$$\mathrm{E}\left( \sum_{n_1, \cdots, n_r \geq 1} a_{n_1, \cdots, n_r} y_{n_1} \cdots y_{n_r} \right) = \sum_{n_1, \cdots, n_r \geq 1} a_{n_1, \cdots, n_r}.$$

*Then we have:*
*(i) For* $r \geq 1$,

$$\mathrm{E}(y_{n_0} * y_{n_1} * \cdots * y_{n_r}) < (r+2)!,$$

*(ii) For* $r \geq 1,\ s \geq 2$,

$$\begin{aligned}
&\mathrm{E}\left(\prod_{i=1}^{s} y_{n_{i0}} * \prod_{i=1}^{s} y_{n_{i1}} * \cdots * \prod_{i=1}^{s} y_{n_{ir}}\right)\\
&\leq \mathrm{E}\left(y_{n_{10}} * \cdots * y_{n_{s0}} * y_{n_{11}} * \cdots * y_{n_{s1}} * \cdots * y_{n_{1r}} * \cdots * y_{n_{sr}}\right)\\
&< [s(r+1)+1]!.
\end{aligned}$$

**Proof.** $(i)$ For $r=1$, one has

$$y_{n_0} * y_{n_1} = y_{n_0}y_{n_1} + y_{n_1}y_{n_0} + y_{n_0+n_1}.$$

There are 2 depth two terms and 1 depth one term in the above expression. Assuming that for $r=p$, there are $B(p,i)$ depth $i+1$ terms in

$$y_{n_0} * y_{n_1} * \cdots * y_{n_p}$$

for $0 \leq i \leq p$. Since

$$\begin{aligned}
&y_m * y_{m_1}y_{m_2}\cdots y_{m_s}\\
&= y_{m_1}y_{m_2}\cdots y_{m_s} * y_m\\
&= y_m y_{m_1}y_{m_2}\cdots y_{m_s} + y_{m_1}y_m y_{m_2}\cdots y_{m_s} + \cdots + y_{m_1}y_{m_2}\cdots y_{m_s}y_m\\
&+ y_{m_1+m}y_{m_2}\cdots y_{m_s} + y_{m_1}y_{m_2+m}\cdots y_{m_s} + \cdots + y_{m_1}y_{m_2}\cdots y_{m_s+m},
\end{aligned}$$

there are $s+1$ depth $s+1$ terms and $s$ depth $s$ terms in the above expression.

As a result, there are $(i+1)B(p,i)+(i+1)B(p,i-1)$ depth $i+1$ terms in

$$y_{n_0} * y_{n_1} * \cdots * y_{n_p} * y_{n_{p+1}}$$

for $0 \leq i \leq p+1$. Thus we have

$$B(p+1,i) = (i+1)B(p,i) + (i+1)B(p,i-1), \tag{2}$$

for $0 \leq i \leq p+1$. Here $B\binom{p}{p+1} = B\binom{p}{-1} = 0$. One can check that

$$B(r,0)=1,\ B(r,r)=(r+1)!,\ \forall\, r \geq 1.$$

From the formula (2), by induction on $i$ and $p$, one can show that

$$B(p+1,i-1) < B(p+1,i),$$

for $1 \leq i \leq p+1$, $p \geq 1$. So

$$\mathrm{E}(y_{n_0} * y_{n_1} * \cdots * y_{n_r}) = \sum_{i=0}^{r} B(r,i) < (r+1)B(r,r) < (r+2)!.$$

$(ii)$ By the definition of stuffle product, it is clear that

$$\prod_{i=1}^{s} y_{n_{i0}} * \prod_{i=1}^{s} y_{n_{i1}} * \cdots * \prod_{i=1}^{s} y_{n_{ir}}$$

is a part of

$$y_{n_{10}} * \cdots * y_{n_{s0}} * y_{n_{11}} * \cdots * y_{n_{s1}} * \cdots * y_{n_{1r}} * \cdots * y_{n_{sr}}.$$

So we have

$$\begin{aligned}
&\mathrm{E}\left(\prod_{i=1}^{s} y_{n_{i0}} * \prod_{i=1}^{s} y_{n_{i1}} * \cdots * \prod_{i=1}^{s} y_{n_{ir}}\right) \\
&\leq \mathrm{E}\left(y_{n_{10}} * \cdots * y_{n_{s0}} * y_{n_{11}} * \cdots * y_{n_{s1}} * \cdots * y_{n_{1r}} * \cdots * y_{n_{sr}}\right) \\
&< [s(r+1)+1]!. \quad \square
\end{aligned}$$

Since the set of any iterated integrals satisfies the shuffle product formula, Theorem 1.2 is equivalent to the following theorem.

**Theorem 4.3.** *For $F = (f_0, f_1)$, where $f_0$ and $f_1$ are continuous functions on $[0,1]$. Define*

$$I_F : \mathfrak{h}^0 \to \mathbb{C}$$

*as*

$$w = u_1 u_2 \cdots u_k \mapsto \int_{0<t_1<t_2<\cdots<t_k<1} \cdots \int \omega_{u_1}(t_1)\omega_{u_2}(t_2)\cdots\omega_{u_k}(t_k),$$

*where $u_1, u_2, \cdots, u_k \in \{x, y\}$ and*

$$\omega_y(t) = f_1(t)dt,\ \omega_x(t) = f_0(t)dt.$$

*If*

$$I_F(u * v) = I_F(u)I_F(v), \forall\, u, v \in \mathfrak{h}^0,$$

*then*

$$I_F(u) = 0, \forall\, u \in \mathfrak{h}^0.$$

**Proof.** Since $f_0$ and $f_1$ are continuous functions on a closed interval, there is an $M$ which satisfies

$$\left|f_0(t)\right| \leq M,\ \left|f_1(t)\right| \leq M,\ \forall\, t \in [0,1].$$

It follows that

$$\left|I_F(u_1 u_2 \cdots u_k)\right| \leq M^k \cdot \int_{0<t_1<t_2<\cdots<t_k<1} \cdots \int dt_1 dt_2 \cdots dt_k = \frac{M^k}{k!},$$

where $u_1, u_2, \cdots, u_k \in \{x, y\}$. Thus the set of iterated $F$-values is bounded.

Denote by

$$c_n = \max_{\substack{k_1,\cdots,k_{r-1}\geq 1, k_r \geq 2 \\ k_1+\cdots k_r \leq n}} \left|I_F(yx^{k_1-1}\cdots yx^{k_r-1})\right|,$$

$$c = \sup_{k_1,\cdots,k_{r-1}\geq 1, k_r \geq 2} \left| I_F(yx^{k_1-1}\cdots yx^{k_r-1}) \right|.$$

Since

$$\lim_{n\to+\infty} \frac{M^n}{n!} = 0,$$

there is a $j \geq 1$ such that

$$\left| I_F(yx^{k_1-1}\cdots yx^{k_r-1}) \right| \leq c_j, \forall\, k_1+\cdots+k_r > j, r \geq 1.$$

As a result, $c = c_n$ for $n > j$. For convenience, we identify $yx^{k_1-1}\cdots yx^{k_r-1}$ with $y_{k_1}y_{k_2}\cdots y_{k_r}$. Assuming that

$$c = \left| I_F(y_{l_1}y_{l_2}\cdots y_{l_s}) \right|,$$

then

$$\begin{aligned}
& c^{r+1} \\
&= \left| I_F\left( \underbrace{y_{l_1}y_{l_2}\cdots y_{l_s} * \cdots * y_{l_1}y_{l_2}\cdots y_{l_s}}_{r+1} \right) \right| \\
&\leq \left| E\left( \underbrace{y_{l_1}y_{l_2}\cdots y_{l_s} * \cdots * y_{l_1}y_{l_2}\cdots y_{l_s}}_{r+1} \right) \right| \frac{M^{(r+1)(l_1+l_2+\cdots+l_s)}}{[(r+1)(l_1+l_2+\cdots+l_s)]!} \\
&< [s(r+1)+1]! \frac{M^{(r+1)(l_1+l_2+\cdots+l_s)}}{[(r+1)(l_1+l_2+\cdots+l_s)]!}.
\end{aligned}$$

Here the last inequality follows from Lemma 4.2. From the above analysis, it follows that

$$c \leq \left( \frac{[s(r+1)+1]!}{[(r+1)(l_1+\cdots+l_s)]!} \right)^{\frac{1}{r+1}} M^{l_1+\cdots+l_s} \leq \left( \frac{[s(r+1)+1]!}{[(r+1)(s+1)]!} \right)^{\frac{1}{r+1}} M^{l_1+\cdots+l_s}.$$

In the above formula, the second inequality follows from

$$l_1, \cdots, l_{s-1} \geq 1, l_s \geq 2.$$

By the Stirling's formula

$$\lim_{n\to+\infty} \frac{n!}{\sqrt{2\pi n}\left(\frac{n}{e}\right)^n} = 1,$$

one has

$$\lim_{r\to+\infty} \left( \frac{[s(r+1)+1]!}{[(r+1)(s+1)]!} \right)^{\frac{1}{r+1}}$$

$$
\begin{aligned}
&= \lim_{r\to+\infty}\left([s(r+1)+1]\frac{[s(r+1)]!}{[(r+1)(s+1)]!}\right)^{\frac{1}{r+1}}\\
&= \lim_{r\to+\infty}\left(\frac{[s(r+1)]!}{[(r+1)(s+1)]!}\right)^{\frac{1}{r+1}}\\
&= \lim_{r\to+\infty}\left(\frac{\sqrt{s(r+1)}\left(\frac{s(r+1)}{e}\right)^{s(r+1)}}{\sqrt{(s+1)(r+1)}\left(\frac{(s+1)(r+1)}{e}\right)^{(s+1)(r+1)}}\right)^{\frac{1}{r+1}}\\
&= \lim_{r\to+\infty}\left(\frac{\left(\frac{s(r+1)}{e}\right)^{s(r+1)}}{\left(\frac{(s+1)(r+1)}{e}\right)^{(s+1)(r+1)}}\right)^{\frac{1}{r+1}}\\
&= \frac{es^s}{(s+1)^{s+1}}\lim_{r\to+\infty}\frac{1}{r+1}\\
&= 0.
\end{aligned}
$$

In conclusion, we have $c = 0$. Thus $I_F(u) = 0, \forall\, u \in \mathfrak{h}^0$. $\square$

**Remark 4.4.** Let $F = (f_0, f_1)$, where $f_0$ and $f_1$ are continuous functions on $[0, 1]$. On one hand, if $f_0 \equiv 0$ or $f_1 \equiv 0$, then it is obvious that

$$I_F(u) = 0, \forall\, u \in \mathfrak{h}^0.$$

On the other hand, from the statement

$$I_F(u) = 0, \forall\, u \in \mathfrak{h}^0,$$

can we deduce that $f_0 \equiv 0$ or $f_1 \equiv 0$?

Let $F = (f_0, f_1)$, where $f_0$ and $f_1$ are continuous $\mathbb{C}$-valued functions on $(0, 1]$ and $[0, 1)$ respectively and

$$|f_1(t)| < \frac{M}{1-t},\ |f_0(t)| < \frac{M}{t}, \forall\, t \in (0, 1).$$

Now we consider the generating series of the iterated $F$-values.

$$
\begin{aligned}
\mathfrak{I}_F(s_1) &= \sum_{k_1\geq 2} I_F(y_{k_1})s_1^{k_1-1},\\
\mathfrak{I}_F(s_1, s_2) &= \sum_{k_1,k_2\geq 2} I_F(y_{k_1}y_{k_2})s_1^{k_1-1}s_2^{k_2-1}.
\end{aligned}
$$

**Proposition 4.5.** *(i) For $\mathfrak{I}_F(s_1)$ and $\mathfrak{I}_F(s_1, s_2)$, we have*

$$\mathfrak{I}_F(s_1) = \int_0^1 f_1(t_1)\left[\exp\left(s_1\int_{t_1}^1 f_0(u)du\right)-1\right]dt_1,$$

$$\mathfrak{I}_F(s_1, s_2) =$$

$$\int_{0<t_1<t_2<1} f_1(t_1)f_1(t_2)\left[\exp\left(s_1\int_{t_1}^{t_2} f_0(u)du\right)-1\right]\left[\exp\left(s_2\int_{t_2}^{1} f_0(v)dv\right)-1\right]dt_1dt_2.$$

*(ii) The following two formulas are equivalent:*

$$I_F(y_{k_1} * y_{k_2}) = I_F(y_{k_1})I_F(y_{k_2}),\ \forall\, k_1, k_2 \geq 2, \tag{a}$$

$$\begin{aligned}&\mathfrak{I}_F(s_1)\mathfrak{I}_F(s_2)\\ &= \mathfrak{I}_F(s_1, s_2) + \mathfrak{I}_F(s_2, s_1) + \frac{\mathfrak{I}_F(s_1) - \mathfrak{I}_F(s_2)}{s_1 - s_2} - \frac{\mathfrak{I}_F(s_1)}{s_1} - \frac{\mathfrak{I}_F(s_2)}{s_2} + I_F(y_2).\end{aligned} \tag{b}$$

**Proof.** $(i)$ follows from the following simple observation

$$\begin{aligned}&\exp\left(s\int_A^B g(t_1)dt_1\right)\\ &= 1 + \int_A^B g(t_1)dt_1 \cdot s + \cdots + \int\limits_{A<t_1<\cdots<t_r<B}\cdots\int g(t_1)\cdots g(t_r)dt_1\cdots dt_r \cdot s^r + \cdots,\end{aligned}$$

where $g(t)$ is a continuous function on $[A, B]$. For example,

$$\begin{aligned}&\mathfrak{I}_F(s_1)\\ &= \sum_{k_1\geq 1} I_F(y_{k_1+1})s_1^{k_1}\\ &= \sum_{k_1\geq 1} s_1^{k_1} \cdot \int\limits_{0<t_1<t_2<\cdots<t_{k_1+1}<1} f_1(t_1)f_0(t_2)\cdots f_0(t_{k_1+1})dt_1dt_2\cdots dt_{k_1+1}\\ &= \sum_{k_1\geq 1} s_1^{k_1}\int_0^1 f_1(t_1)\frac{\left(\int_{t_1}^1 f_0(u)du\right)^{k_1}}{k_1!}dt_1\\ &= \int_0^1 f_1(t_1)\sum_{k_1\geq 1}\frac{\left(\int_{t_1}^1 f_0(u)du\right)^{k_1}}{k_1!}dt_1\\ &= \int_0^1 f_1(t_1)\left[\exp\left(s_1\int_{t_1}^1 f_0(u)du\right) - 1\right]dt_1.\end{aligned}$$

$(ii)$ By definition, the statement $(a)$ is equivalent to

$$I_F(y_{k_1}y_{k_2}) + I_F(y_{k_2}y_{k_1}) + I_F(y_{k_1+k_2}) = I_F(k_1)I_F(k_2),\ \forall\, k_1, k_2 \geq 2. \tag{3}$$

The condition (3) is equivalent to

$$\sum_{k_1,k_2\geq 2}\left(I_F(y_{k_1}y_{k_2}) + I_F(y_{k_2}y_{k_1}) + I_F(y_{k_1+k_2})\right)s_1^{k_1-1}s_2^{k_2-1} = \mathfrak{I}_F(s_1)\mathfrak{I}_F(s_2).$$

By definition, the left hand side is

$$\begin{aligned}
&\sum_{k_1,k_2\geq 2}\left(I_F(y_{k_1}y_{k_2})+I_F(y_{k_2}y_{k_1})+I_F(y_{k_1+k_2})\right)s_1^{k_1-1}s_2^{k_2-1}\\
&=\mathfrak{I}_F(s_1,s_2)+\mathfrak{I}_F(s_2,s_1)+\sum_{k\geq 4}I_F(y_k)\sum_{\substack{k_1+k_2=k\\ k_1,k_2\geq 2}}s_1^{k_1-1}s_2^{k_2-1}\\
&=\mathfrak{I}_F(s_1,s_2)+\mathfrak{I}_F(s_2,s_1)+\sum_{k\geq 4}I_F(y_k)\left(\frac{s_1^{k-1}-s_2^{k-1}}{s_1-s_2}-s_1^{k-2}-s_2^{k-2}\right)\\
&=\mathfrak{I}_F(s_1,s_2)+\mathfrak{I}_F(s_2,s_1)+\frac{\mathfrak{I}_F(s_1)-\mathfrak{I}_F(s_2)}{s_1-s_2}-\frac{\mathfrak{I}_F(s_1)}{s_1}-\frac{\mathfrak{I}_F(s_2)}{s_2}+I_F(y_2).
\end{aligned}$$

As a result, the statement $(ii)$ is proved. □

For $F=(f_0,f_1)=\left(\frac{1}{t},\frac{1}{1-t}\right)$, by Proposition 4.5, one has

$$\mathfrak{I}_F(s_1)=\int_0^1\frac{1}{1-t_1}\left[\left(\frac{1}{t_1}\right)^{s_1}-1\right]dt_1,$$

$$\mathfrak{I}_F(s_1,s_2)=\int\limits_{0<t_1<t_2<1}\frac{1}{(1-t_1)(1-t_2)}\left[\left(\frac{t_2}{t_1}\right)^{s_1}-1\right]\left[\left(\frac{1}{t_2}\right)^{s_2}-1\right]dt_1dt_2.$$

By the stuffle product of multiple zeta values, the functional equation $(b)$ holds for the above $\mathfrak{I}_F(s_1)$ and $\mathfrak{I}_F(s_1,s_2)$. Here we give another proof of the functional equation $(b)$ for $F=(f_0,f_1)=\left(\frac{1}{t},\frac{1}{1-t}\right)$.

**Proposition 4.6.** *For*

$$\mathfrak{I}_F(s_1)=\int_0^1\frac{1}{1-t_1}\left[\left(\frac{1}{t_1}\right)^{s_1}-1\right]dt_1,$$

$$\mathfrak{I}_F(s_1,s_2)=\int\limits_{0<t_1<t_2<1}\frac{1}{(1-t_1)(1-t_2)}\left[\left(\frac{t_2}{t_1}\right)^{s_1}-1\right]\left[\left(\frac{1}{t_2}\right)^{s_2}-1\right]dt_1dt_2,$$

*one has*

$$\begin{aligned}
&\mathfrak{I}_F(s_1)\mathfrak{I}_F(s_2)\\
&=\mathfrak{I}_F(s_1,s_2)+\mathfrak{I}_F(s_2,s_1)+\frac{\mathfrak{I}_F(s_1)-\mathfrak{I}_F(s_2)}{s_1-s_2}-\frac{\mathfrak{I}_F(s_1)}{s_1}-\frac{\mathfrak{I}_F(s_2)}{s_2}+\zeta(2).
\end{aligned}$$

**Proof.** It is easy to check that

$$\frac{1}{(1-t_1)(1-t_2)}=\frac{1}{(1-t_1)(1-t_1t_2)}+\frac{1}{(1-t_2)(1-t_1t_2)}-\frac{1}{1-t_1t_2}.$$

Thus we have

$$\begin{aligned}
&\frac{1}{(1-t_1)(1-t_2)}\left[\left(\frac{1}{t_1}\right)^{s_1}-1\right]\left[\left(\frac{1}{t_2}\right)^{s_2}-1\right]\\
&=\left[\frac{1}{(1-t_1)(1-t_1t_2)}+\frac{1}{(1-t_2)(1-t_1t_2)}-\frac{1}{1-t_1t_2}\right]\left[\left(\frac{1}{t_1}\right)^{s_1}-1\right]\left[\left(\frac{1}{t_2}\right)^{s_2}-1\right].
\end{aligned}$$

It follows that

$$\begin{aligned}
&\mathfrak{I}_F(s_1)\mathfrak{I}_F(s_2)\\
&=\iint\limits_{[0,1]^2}\frac{1}{(1-t_1)(1-t_2)}\left[\left(\frac{1}{t_1}\right)^{s_1}-1\right]\left[\left(\frac{1}{t_2}\right)^{s_2}-1\right]dt_1dt_2\\
&=I_1+I_2-I_3.
\end{aligned}$$

Here

$$\begin{aligned}
&I_1\\
&=\iint\limits_{[0,1]^2}\frac{1}{(1-t_1)(1-t_1t_2)}\left[\left(\frac{1}{t_1}\right)^{s_1}-1\right]\left[\left(\frac{1}{t_2}\right)^{s_2}-1\right]dt_1dt_2\\
&=\int\limits_{1>t_1>t_2>0}\frac{1}{(1-t_1)(1-t_2)}\left[\left(\frac{1}{t_1}\right)^{s_1}-1\right]\left[\left(\frac{t_1}{t_2}\right)^{s_2}-1\right]\frac{dt_1dt_2}{t_1}\\
&=\int\limits_{0<t_1<t_2<1}\frac{1}{(1-t_1)(1-t_2)}\left[\left(\frac{t_2}{t_1}\right)^{s_2}-1\right]\left[\left(\frac{1}{t_2}\right)^{s_1}-1\right]\frac{dt_1dt_2}{t_2}\\
&=\int\limits_{0<t_1<t_2<1}\frac{1}{1-t_1}\left[\left(\frac{t_2}{t_1}\right)^{s_2}-1\right]\left[\left(\frac{1}{t_2}\right)^{s_1}-1\right]\left(\frac{1}{1-t_2}+\frac{1}{t_2}\right)dt_1dt_2,\\
&=\int\limits_{0<t_1<t_2<1}\frac{1}{(1-t_1)(1-t_2)}\left[\left(\frac{t_2}{t_1}\right)^{s_2}-1\right]\left[\left(\frac{1}{t_2}\right)^{s_1}-1\right]dt_1dt_2\\
&+\int\limits_{0<t_1<t_2<1}\frac{1}{(1-t_1)t_2}\left[\left(\frac{t_2}{t_1}\right)^{s_2}-1\right]\left[\left(\frac{1}{t_2}\right)^{s_1}-1\right]dt_1dt_2,
\end{aligned}$$

$$\begin{aligned}
&I_2\\
&=\iint\limits_{[0,1]^2}\frac{1}{(1-t_2)(1-t_1t_2)}\left[\left(\frac{1}{t_1}\right)^{s_1}-1\right]\left[\left(\frac{1}{t_2}\right)^{s_2}-1\right]dt_1dt_2\\
&=\int\limits_{0<t_1<t_2<1}\frac{1}{(1-t_1)(1-t_2)}\left[\left(\frac{t_2}{t_1}\right)^{s_1}-1\right]\left[\left(\frac{1}{t_2}\right)^{s_2}-1\right]\frac{dt_1dt_2}{t_2}\\
&=\int\limits_{0<t_1<t_2<1}\frac{1}{1-t_1}\left[\left(\frac{t_2}{t_1}\right)^{s_1}-1\right]\left[\left(\frac{1}{t_2}\right)^{s_2}-1\right]\left(\frac{1}{1-t_2}+\frac{1}{t_2}\right)dt_1dt_2\\
&=\int\limits_{0<t_1<t_2<1}\frac{1}{(1-t_1)(1-t_2)}\left[\left(\frac{t_2}{t_1}\right)^{s_1}-1\right]\left[\left(\frac{1}{t_2}\right)^{s_2}-1\right]dt_1dt_2\\
&+\int\limits_{0<t_1<t_2<1}\frac{1}{(1-t_1)t_2}\left[\left(\frac{t_2}{t_1}\right)^{s_1}-1\right]\left[\left(\frac{1}{t_2}\right)^{s_2}-1\right]dt_1dt_2,
\end{aligned}$$

$$
\begin{aligned}
&I_3\\
&= \iint\limits_{[0,1]^2} \frac{1}{1-t_1t_2}\left[\left(\frac{1}{t_1}\right)^{s_1}-1\right]\left[\left(\frac{1}{t_2}\right)^{s_2}-1\right]dt_1dt_2\\
&= \int\limits_{0<t_1<t_2<1} \frac{1}{1-t_1}\left[\left(\frac{t_2}{t_1}\right)^{s_1}-1\right]\left[\left(\frac{1}{t_2}\right)^{s_2}-1\right]\frac{dt_1dt_2}{t_2}\\
&= \int\limits_{0<t_1<t_2<1} \frac{1}{(1-t_1)t_2}\left[\left(\frac{t_2}{t_1}\right)^{s_1}-1\right]\left[\left(\frac{1}{t_2}\right)^{s_2}-1\right]dt_1dt_2.
\end{aligned}
$$

By the above calculations, we have

$$
\begin{aligned}
&\mathfrak{I}_F(s_1)\mathfrak{I}_F(s_2)\\
&= I_1+I_2-I_3\\
&= \int\limits_{0<t_1<t_2<1} \frac{1}{(1-t_1)(1-t_2)}\left[\left(\frac{t_2}{t_1}\right)^{s_2}-1\right]\left[\left(\frac{1}{t_2}\right)^{s_1}-1\right]dt_1dt_2\\
&+ \int\limits_{0<t_1<t_2<1} \frac{1}{(1-t_1)(1-t_2)}\left[\left(\frac{t_2}{t_1}\right)^{s_1}-1\right]\left[\left(\frac{1}{t_2}\right)^{s_2}-1\right]dt_1dt_2\\
&+ \int\limits_{0<t_1<t_2<1} \frac{1}{(1-t_1)t_2}\left[\left(\frac{t_2}{t_1}\right)^{s_2}-1\right]\left[\left(\frac{1}{t_2}\right)^{s_1}-1\right]dt_1dt_2
\end{aligned}
$$

As a result, it suffices to shows that

$$
\begin{aligned}
&\int\limits_{0<t_1<t_2<1} \frac{1}{(1-t_1)t_2}\left[\left(\frac{t_2}{t_1}\right)^{s_2}-1\right]\left[\left(\frac{1}{t_2}\right)^{s_1}-1\right]dt_1dt_2\\
&= \frac{\mathfrak{I}_F(s_1)-\mathfrak{I}_F(s_2)}{s_1-s_2}-\frac{\mathfrak{I}_F(s_1)}{s_1}-\frac{\mathfrak{I}_F(s_2)}{s_2}+\zeta(2).
\end{aligned}
$$

The above formula follows from integration on the variable $t_2$, i. e.

$$
\begin{aligned}
&\int\limits_{0<t_1<t_2<1} \frac{1}{(1-t_1)t_2}\left[\left(\frac{t_2}{t_1}\right)^{s_2}-1\right]\left[\left(\frac{1}{t_2}\right)^{s_1}-1\right]dt_1dt_2\\
&= \int\limits_{0<t_1<t_2<1} \frac{1}{1-t_1}\left(t_1^{-s_2}t_2^{s_2-s_1-1}-t_1^{-s_2}t_2^{s_2-1}-t_2^{-s_1-1}\right)dt_1dt_2+\zeta(2)\\
&= \int\limits_{0<t_1<1} \frac{1}{1-t_1}\left[t_1^{-s_2}\left(\frac{t_2^{s_2-s_1}}{s_2-s_1}\Bigg|_{t_2=t_1}^{t_2=1}\right)-t_1^{-s_2}\left(\frac{t_2^{s_2}}{s_2}\Bigg|_{t_2=t_1}^{t_2=1}\right)+\left(\frac{t_2^{-s_1}}{s_1}\Bigg|_{t_2=t_1}^{t_2=1}\right)\right]dt_1+\zeta(2)\\
&= \int\limits_{0<t_1<1} \frac{1}{1-t_1}\left[\frac{t_1^{-s_2}-t_1^{-s_1}}{s_2-s_1}-\frac{t_1^{-s_2}-1}{s_2}-\frac{t_1^{-s_1}-1}{s_1}\right]dt_1+\zeta(2).\\
&= \frac{\mathfrak{I}_F(s_1)-\mathfrak{I}_F(s_2)}{s_1-s_2}-\frac{\mathfrak{I}_F(s_1)}{s_1}-\frac{\mathfrak{I}_F(s_2)}{s_2}+\zeta(2). \quad \square
\end{aligned}
$$

For $k_1, \cdots, k_r \geq 1$, the one variable multiple polylogarithm $\mathrm{Li}_{k_1,\cdots,k_r}(z)$ can be written as

$$\mathrm{Li}_{k_1,\cdots,k_r}(z) = \int\cdots\int_{0<t_1<\cdots<t_k<z} \omega_1(t_1)\cdots\omega_k(t_k).$$

Here $k = k_1 + \cdots + k_r$ and

$$\omega_i(t) = \begin{cases} \frac{1}{1-t}, & i \in \{1, 1+k_1, \cdots, 1+k_1+\cdots+k_{r-1}\}; \\ \frac{1}{t}, & i \notin \{1, 1+k_1, \cdots, 1+k_1+\cdots+k_{r-1}\}. \end{cases}$$

From the above iterated integral representation, it is clear that the set of one variable multiple polylogarithms

$$\left\{\mathrm{Li}_{k_1,\cdots,k_r}(z) \,\middle|\, k_1, \cdots, k_{r-1} \geq 1, k_r \geq 2\right\}$$

satisfies the shuffle product formula. Thus Theorem 1.3 is equivalent to the following theorem.

**Theorem 4.7.** *For $|z| \leq 1$, define $I_z : \mathfrak{h}^0 \to \mathbb{C}$ by*

$$I_z(y_{k_1}\cdots y_{k_r}) = \mathrm{Li}_{k_1,\cdots,k_r}(z),\ \forall\, k_1, \cdots, k_{r-1} \geq 1, k_r \geq 2.$$

*If*

$$I_z(u * v) = I_z(u)I_z(v),\ \forall\, u, v \in \mathfrak{h}^0,$$

*then $z = 0$ or $1$.*

**Proof.** Denote by

$$\mathfrak{I}_z(s) = \sum_{k\geq 2} \mathrm{Li}_k(z)s^{k-1},\ \mathfrak{I}_z(s_1, s_2) = \sum_{k_1,k_2\geq 2} \mathrm{Li}_{k_1,k_2}(z)s_1^{k_1-1}s_2^{k_2-1}.$$

One can check that

$$\mathfrak{I}_z(s) = \sum_{n\geq 1} \frac{s}{n(n-s)} \cdot z^n,\ \mathfrak{I}_z(s_1, s_2) = \sum_{1\leq n_1<n_2} \frac{s_1 s_2}{n_1 n_2(n_1 - s_1)(n_2 - s_2)} \cdot z^{n_2}.$$

By Proposition 4.5 $(ii)$, one has

$$\begin{aligned} &\sum_{n_1,n_2\geq 1} \frac{s_1 s_1}{n_1 n_2(n_1 - s_1)(n_2 - s_2)} \cdot z^{n_1+n_2} \\ =& \sum_{1\leq n_1<n_2} \frac{s_1 s_2}{n_1 n_2(n_1 - s_1)(n_2 - s_2)} \cdot z^{n_2} + \sum_{1\leq n_1<n_2} \frac{s_1 s_2}{n_1 n_2(n_1 - s_1)(n_2 - s_2)} \cdot z^{n_2} \\ &- \sum_{n\geq 1} \frac{1}{(n-s_1)(n-s_2)} \cdot z^n - \sum_{n\geq 1} \frac{1}{n(n-s_1)} z^n - \sum_{n\geq 1} \frac{1}{n(n-s_2)} z^n + \sum_{n\geq 1} \frac{z^n}{n^2}. \end{aligned}$$

In the above formula, the left hand side and the right hand side are both meromorphic functions of variables $(s_1, s_2)$ over $\mathbb{C}^2$. Thus for $1 \leq m_1 < m_2$,

$$
\begin{aligned}
&z^{m_1+m_2}\\
&=\lim_{(s_1,s_2)\to(m_1,m_2)}(m_1-s_1)(m_2-s_2)\sum_{n_1,n_2\geq 1}\frac{s_1s_1}{n_1n_2(n_1-s_1)(n_2-s_2)}\cdot z^{n_1+n_2}\\
&=\lim_{(s_1,s_2)\to(m_1,m_2)}(m_1-s_1)(m_2-s_2)\Bigg\{\sum_{1\leq n_1<n_2}\frac{s_1s_2}{n_1n_2(n_1-s_1)(n_2-s_2)}\cdot z^{n_2}\\
&+\sum_{1\leq n_1<n_2}\frac{s_1s_2}{n_1n_2(n_1-s_1)(n_2-s_2)}\cdot z^{n_2}-\sum_{n\geq 1}\frac{1}{(n-s_1)(n-s_2)}\cdot z^n-\sum_{n\geq 1}\frac{1}{n(n-s_1)}z^n\\
&-\sum_{n\geq 1}\frac{1}{n(n-s_2)}z^n+\sum_{n\geq 1}\frac{z^n}{n^2}\Bigg\}\\
&=z^{m_2}.
\end{aligned}
$$

Since the above formula holds for any $1\leq m_1<m_2$, one has $z=0$ or 1. $\square$

**Remark 4.8.** For the following generalization of one variable multiple poloylogarithms:

$$\mathcal{L}_{k_1,\cdots,k_r}(z)=\int\cdots\int_{0<t_1<\cdots<t_k<z}\omega_1(t_1)\cdots\omega_k(t_k),$$

where $k=k_1+\cdots+k_r$ and

$$|\alpha|\leq 1,\beta>0,$$

$$\omega_i(t)=\begin{cases}\frac{1}{1-\alpha\cdot t^\beta}, & i\in\{1,1+k_1,\cdots,1+k_1+\cdots+k_{r-1}\};\\ \frac{1}{t}, & i\notin\{1,1+k_1,\cdots,1+k_1+\cdots+k_{r-1}\},\end{cases}$$

one can prove the same result as Theorem 4.7 by the same method.

**Remark 4.9.** If the multiple $G$-values or the iterated $F$-values are divergent, we may need the theory of renormalization in [6] to investigate their structures.

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